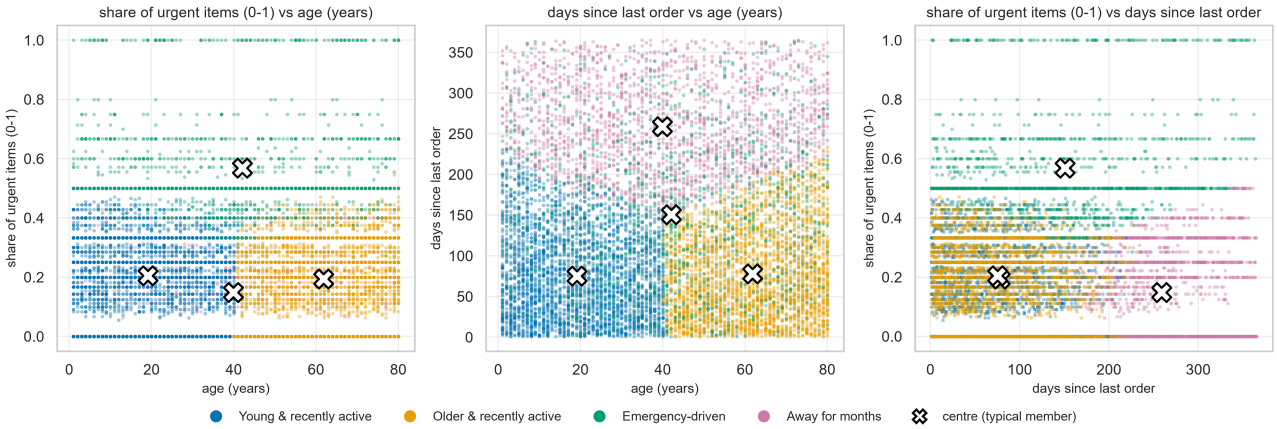
Patient Segmentation - K-Means Clustering Results

- The laboratory's 9,776 patients were grouped into 4 segments with K-means.
- Clustering features: days since last order, patient age, and the share of each patient's items booked as stat/emergency
- standardised before clustering.
- The chart shows the same patients three times, each panel hiding one feature (age vs urgency, age vs last visit, last visit vs urgency). Every dot is a patient; the colour is their segment; the white X is the segment's typical member.



Key insights

- Four segments by cleanest separation (k = 4, stable across four random restarts): Young & recently active (3,124), Older & recently active (3,138), Away for months (2,111), Emergency-driven (1,403).
- The two active segments are twins: same visiting rhythm (~2.5 months since the last order), same urgency mix and spend - only age differs (~19 vs ~62). They are 64% of patients and ~73% of the 54.5M USD revenue.
- Emergency-driven (~14% of patients) books most of its tests as stat/emergency (~57% of their items) - 11% of revenue,
- Away for months (22%) last visited ~8.5 months ago, with the fewest orders and lowest spend
- The dense line at the bottom of the urgency panels is 22% of patients who never order an urgent test

Actions

- Plan the fast lane around the Emergency-driven group - it is their urgent work that keeps the lane busy, not the total number of patients.
- Run a campaign campaign targeting the Away-for-months segment as it is large and cheap to target.